Violeta Kastreva

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Research Interests: Representation Learning, Information Retrieval, LLM Evaluation, Mechanistic Interpretability.

Education

ETH Zurich — Zurich, Switzerland
MSc in Computer Science | Major: Machine Intelligence, Minor: Theoretical Computer Science — 09/2026 – 07/2028

Sofia University — Sofia, Bulgaria
BSc in Computer Science | **GPA: 5.88/6.00** (Top 1%) — 10/2022 – 06/2026

Publications

Tokenisation over Bounded Alphabets is Hard — Page link
V. Kastreva*, P. Whittington*, D. Komm, T. Pimentel — arXiv link
ICLR 2026

How Correct Is Your Answer? A Semantic Correctness Framework for Open QA Evaluation — Page link
E. Yotkova*, **V. Kastreva***, P. Velkov, H. Boyanov, D. Dimitrov, P. Nakov, I. Koychev — arXiv link
***EMNLP 2026** Findings*

Lightweight Detection of LLM-Generated Code via Stylometric Signals — ACL Anthology link
E. Yotkova*, **V. Kastreva***, D. Dimitrov, I. Koychev, P. Nakov
*SemEval (**ACL 2026** workshop)*

Byte n-gram Encoding
M. Tomek, P. Whittington, **V. Kastreva**, C. Meister, D. Komm, T. Pimentel
*TokShop (**COLM 2026** workshop)*

Experience

Research Fellow — 09/2026 – 12/2026
SPAR - Research Program for AI Risks — Zurich, Switzerland
- Investigating function-preserving and data-dependent model symmetries as a tool for mechanistic interpretability.

Quantitative Trading Intern — 06/2026 – 08/2026
IMC Trading — Amsterdam, Netherlands
- Developed machine-learning signals for systematic trading in Indian equities using large-scale market data.

Research Assistant — 11/2025 – 06/2026
Sofia University — Sofia, Bulgaria
- Studied cross-seed representation alignment and downstream transfer in language models (paper in preparation).
- Developed a **LLM-generated code detector** using stylometric features. Work accepted to an **ACL 2026** workshop.
- Worked on Open-VEQA, a multilingual and multimodal benchmark for evaluating open-ended visual QA.

Research Fellow — 07/2025 – 09/2025
ETH Zurich — Zurich, Switzerland
- Studied LLM tokenisation under alternative objectives, comparing entropy- and length-based formulations.
- Proved NP-completeness of optimal direct and bottom-up tokenisation; work accepted to **ICLR 2026**.
- Proved limitations of greedy BPE-style tokenisation, showing an unbounded approximation ratio over unary alphabets.

Teaching Assistant — 10/2023 – 06/2026
Sofia University — Sofia, Bulgaria
- Taught exercise sessions for 8 courses, incl. Calculus, Linear Algebra, Probability & Statistics, Theoretical CS, C++.
- Prepared the university mathematics team for **international competitions**, designing 30+ problem sets and exams.

Software Engineer — 11/2023 – 03/2025
Bulgarian National Television — Sofia, Bulgaria
- Developed a high-efficiency media search engine combining keyword and semantic methods with sub-50ms latency.
- Built a two-stage recommendation system (candidate generation + re-ranking), boosting engagement by 15%.

Software Engineer Intern — 06/2023 – 09/2023
DXC Technology — Sofia, Bulgaria
- Chosen as a team lead of 10 intern cohort.

Honors and Awards

Student of the Year - Mathematics and Informatics 2024, 2026 — 05/2024, 05/2026

Google Generation Scholarship (EMEA) — 10/2025
- Awarded for outstanding performance in Computer Science and Mathematics across EMEA.

Silver Medal - Vojtěch Jarník International Mathematical Competition — 05/2025

Student of the Year 2025 — 05/2025
- Chosen out of 15000+ candidates, including PhD and MSc students, to receive the title Best Performing Student.

Silver Medal - Open Mathematical International Olympiad for University Students (OMOUS) — 04/2024
- Competed amongst 600+ students from over 20 countries.

Gold Medal - Team Open International Mathematical Olympiad for University Students — 04/2024

First-place Gold Medal - National Mathematics Competition — 12/2023